AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Type 1: Research Project

ReXTrust: XAI-Driven Multi-Disease Diagnosis in Chest X-rays via Conditional Enhancement and Grounded Reporting

Team Number: 95

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Problem Statement

Chest X-ray diagnosis using AI suffers from three unresolved challenges:

- **Degraded Image Quality**
  X-rays from low-resource settings suffer from noise, low contrast, and poor illumination, reducing diagnostic reliability

- **Lack of Per-Disease Spatial Verification**
  Existing explainability methods produce blended saliency maps, making it difficult to localize individual diseases

- **Ungrounded AI-Generated Reports**
  Vision-language models generate fluent clinical text without verification, leading to potential hallucinated diagnoses

Objective

To build one pipeline that takes a raw chest X-ray and produces a diagnosis you can actually trust:

- **Enhance only when needed**
  To automatically judge whether an X-ray is already clear enough, and only improve the ones that genuinely need it — without losing important medical details.
- **Pick the right detection model, fairly**
  Test CNN, Transformer, and graph-based models under the exact same conditions, so the best model is chosen because it's genuinely better — not because it got an easier setup.
- **Show proof for every disease, separately**
  For each disease the model predicts, show exactly where in the image that prediction came from — one clear picture per disease, not one mixed map.
- **Write a report, then check it**
  Generate a radiology report in plain language, then verify every claim in it before treating it as final.

Research Gaps We Address

- **Enhancement is one-size-fits-all.**
  Existing methods enhance every image the same way, whether it needs it or not, and rarely check if small but important details — like tubes or wires — are still visible afterward.
- **One heatmap has to explain everything.**
  When a model predicts several diseases at once, most tools give one combined heatmap, so there's no way to tell which part of the image actually supports which diagnosis.
- **AI reports aren't fact-checked properly.**
  Report-writing models can sound confident while being wrong, and most checking methods either verify every line the same way, or don't verify anything at all.

Architecture

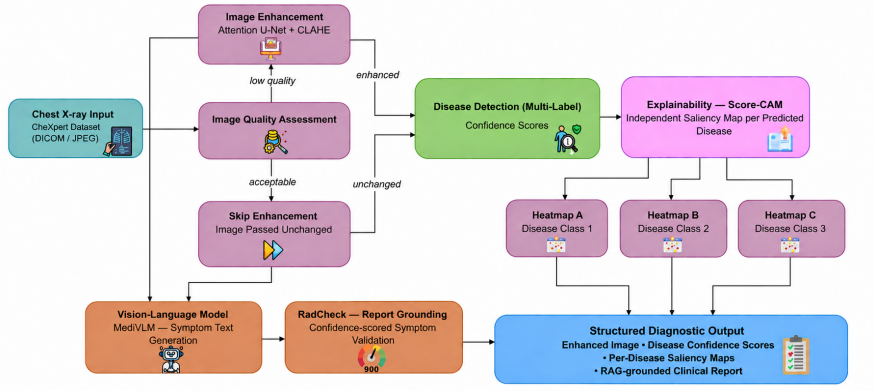


Image Enhancement

Where this module fits in the complete final-year project



Problem Chest X-rays can contain low contrast, noise and weak local structure; enhancement must improve visibility without introducing artifacts that affect downstream disease analysis.

Objective Develop an enhancement model that improves visual quality and contrast of chest X-rays while preserving anatomical detail for the subsequent detection and reporting stages.

Current focus: Integrating the Image Enhancement module into the complete pipeline

Methodology — Enhancement Methods Implemented & Compared

Classical & Learned Baselines

- **CLAHE** — clip-limited adaptive histogram equalization (classical baseline)
- **UniMIE** — pretrained diffusion prior (guided-diffusion backbone), Universal Medical Image Enhancement (UniMIE) [Fei et al., 2025]
- **Zero-DCE** — unsupervised DCE-Net; pixel-wise curve estimation via spatial-consistency, exposure, color-constancy & illumination-smoothness losses [Guo et al., 2020]
- **U-Net** — supervised, trained on CLAHE-target pairs with L1 + SSIM + Laplacian loss (60 epochs) [Ronneberger et al., 2015]

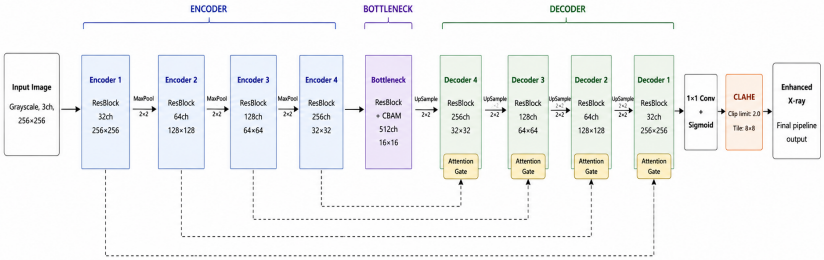
Proposed Hybrid Pipelines

- **U-Net + CLAHE** — U-Net denoises, CLAHE then boosts contrast on the clean output
- **Attention U-Net + CBAM** — 4-stage ResBlock encoder/decoder, CBAM channel+spatial attention at the bottleneck, Attention Gates on every skip connection
- **Attention U-Net + CLAHE (Ours)** — final proposed pipeline; best average rank across all 5 evaluation metrics

Evaluation Metrics

- BRISQUE, Entropy, Sharpness, CII, EME — on 23 low-quality CheXpert images

Proposed Attention U-Net + CLAHE



Residual Blocks

Helps preserve important features while improving feature flow through the network.

CBAM Focuses on important features by applying channel and spatial attention at the bottleneck.

Attention Gates

Selectively pass relevant encoder features to the decoder through the skip connections.

CLAHE Enhances local contrast while limiting excessive amplification of image noise.

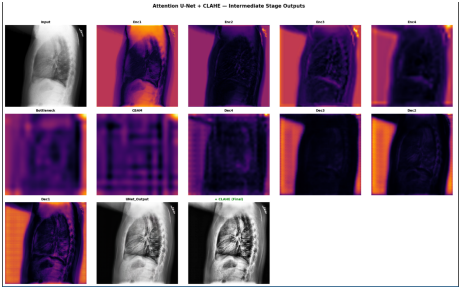
Experimental Results & Qualitative Comparison

Quantitative Leaderboard (23 images)

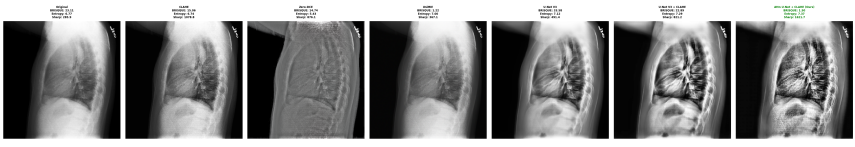
Final Evaluation — 23 low-quality CheXpert images, 8 methods, 5 metrics
(mean values; lower BRISQUE / higher Entropy, Sharpness, CII are better)

Method	BRISQUE↓	Entropy↑	Sharpness↑	CII	Avg Rank
Attn U-Net +	**13.35**	**7.92**	**2082.3**	**1.90**	**2.2**
U-Net V3 + CL	17.51	7.91	1859.7	1.76	3.2
CLAHE	7.01	7.84	3609.6	1.71	3.8
Attention U-Ne	13.22	7.86	1295.7	1.37	4.6
Original	5.62	7.85	534.3	1.00	5.2
Zero-DCE	19.59	6.54	1766.4	1.27	5.4
U-Net V3	17.96	7.86	1190.9	1.29	5.6
UniMIE	-0.73	7.85	725.5	0.93	6.0

Layer-wise Intermediate Feature Maps — Attention U-Net

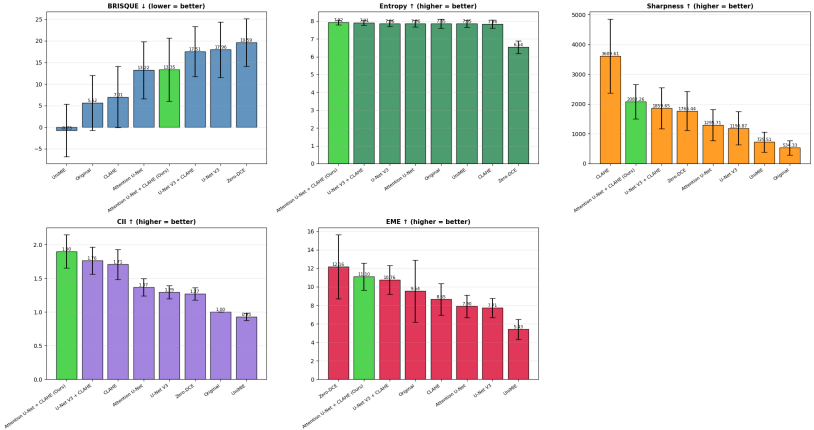


Qualitative Comparison — All Methods, Same Input Image



Quantitative Metric Comparison — All 8 Methods (23 Images)

Mean ± Std per metric, all 8 methods



Attn U-Net + CLAHE ranks top-2 on 4 of 5 metrics, with the highest Entropy and highest CII of all 8 methods — confirming it retains the most diagnostically-relevant structure while enhancing contrast.

Key Takeaways

2.2

Best Avg. Rank

Across BRISQUE, Entropy, Sharpness, CII & EME — 23 CheXpert images, 8 methods compared

7.92

Highest Entropy

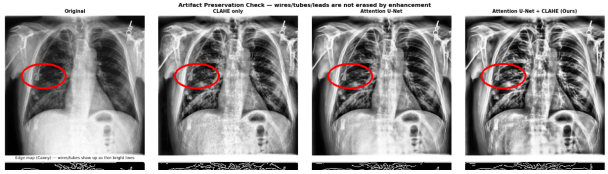
Retains the most diagnostically-relevant structural detail of all 8 methods tested

1.90

Highest CII

Best contrast improvement of all 8 methods — clearer anatomical boundaries

- **Consistent -** the barplots (previous slide) show it stays in the top 2 methods on 4 of 5 metrics across all the 23 test images



Original vs. CLAHE-only vs. Attention U-Net vs. Attention U-Net + CLAHE

Clinical Detail Preserved

Support device **(circled)** remains visible even after enhancement.

Quality improves without erasing clinically relevant structural detail.

Disease Detection



Problem

Multi-label chest X-ray classification must not only predict disease probabilities, but also let each prediction be independently, spatially verified — a single blended saliency map is not enough.

Objective

Predict Cardiomegaly, Edema, and Pleural Effusion from chest X-rays, and compare architectures under one identical, leak-free data and evaluation protocol so gains are attributable to design, not luck.

Iterative Progress

CNN (baseline recipe) → CNN (matched recipe, ablation) → Hybrid CNN-Transformer → CheXGCN (label co-occurrence GNN).

Current focus: comparing all 4 techniques under one shared protocol.

Current focus: comparing all 4 techniques under one shared evaluation protocol

Shared Data Foundation

Stage 0 — built once, reused identically by all 4 Disease Detection techniques



Problem

Comparing 4 architectures is only meaningful if every technique sees identical data. Any difference in sampling, splitting, or preprocessing would confound the comparison, rather than reflecting a true architectural difference.

Approach

A single, fixed pipeline is built once and reused by all four techniques: patient-level split (no patient's scans span multiple splits), balanced per-disease subsampling with a genuine negative pool, and one shared loader / train / evaluate function.

Guarantee

Every technique trains, validates, and is tested on the exact same images, with thresholds tuned only on validation data. Architecture remains the only variable in the model's comparison.

One fixed data pipeline, reused identically across all 4 techniques — no per-technique variation

Tested Multi-Technique Disease Detection Pipeline

All four techniques train, validate, and test on the same patient-level split, data loaders, and evaluation functions — thresholds are tuned on validation data only.

1. CNN (Baseline Recipe)

DenseNet121 $\rightarrow$ GAP $\rightarrow$ Linear(3). Plain BCE loss, no dropout, no scheduler — the literal CheXNet recipe, used as a literature-comparable reference.

2. CNN (Matched Recipe)

Same architecture as the baseline. Focal Loss + dropout (0.3) + LR scheduler — isolates the recipe effect, and matches the recipe used by techniques 3 and 4.

Tested Multi-Technique Disease Detection Pipeline

3. Hybrid CNN-Transformer

DenseNet121 features → patch tokens + [CLS] → 2-layer Transformer encoder → classify off [CLS]. Uses the same matched recipe as technique 2.

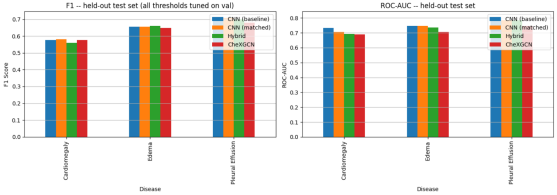
4. CheXGCN

Image features + a GCN over the disease co-occurrence graph (built from the train split only). Dot-product classifier. Uses the same matched recipe as technique 2.

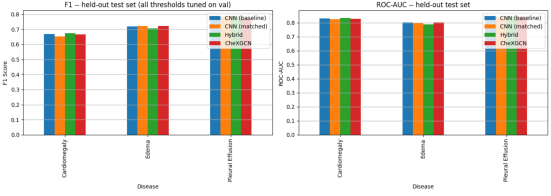
- Thresholds are tuned per disease on `val_loader` only, then frozen and applied to `test_loader` for Precision/Recall/F1.

Evaluation Comparison Across Data Scales

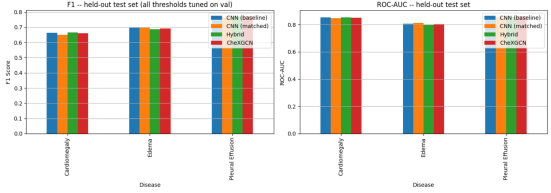
1k per disease



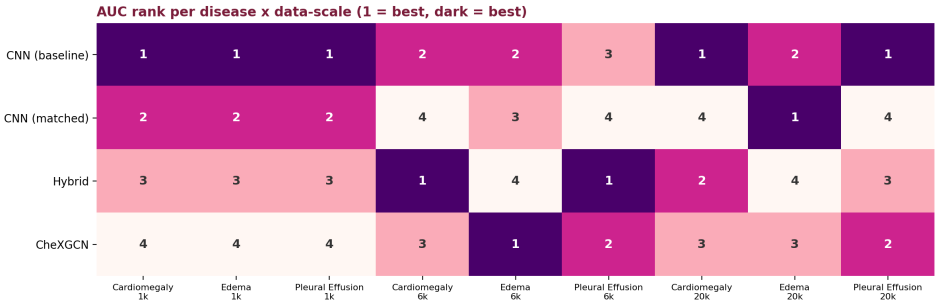
6k per disease

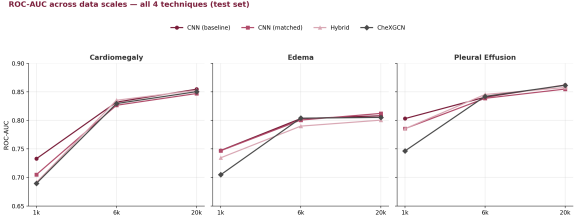


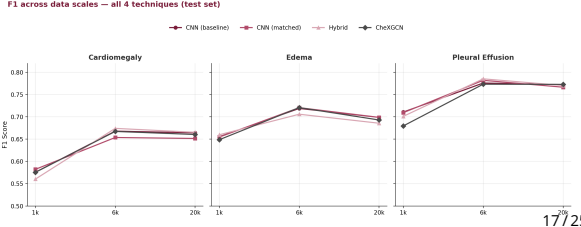
20k per disease



Evaluation Comparison Across Data Scales







Key Takeaways

- **CheXGCN shows the largest improvement with increased data.** It records the lowest AUC at 1k but is tied for the highest AUC on Cardiomegaly and Pleural Effusion at 20k — consistent with the graph-based label co-occurrence signal requiring a larger sample size to be reliable.
- **Most performance gains occur between 1k and 6k images.** All four techniques show a sharp increase in AUC/F1 from 1k to 6k, followed by a plateau (and in some cases a slight decrease) from 6k to 20k, indicating diminishing returns beyond the 6k scale.
- **The Hybrid CNN-Transformer requires more data to perform competitively.** It records the lowest or near-lowest F1/AUC at 1k, consistent with the higher data requirements typically associated with Transformer-based attention compared to the built-in spatial inductive bias of convolutional architectures.
- **Relative rankings change substantially as data scale increases.** At 6k and 20k, no single architecture leads consistently — Hybrid and CheXGCN each achieve the top rank for at least one disease, while CNN (baseline) drops to second or third place on several occasions. Architecture choice has less impact than data volume at this scale.

Score-CAM on DenseNet121



Why Score-CAM

Gradient-free: instead of backpropagating gradients (Grad-CAM), each activation map is used directly as an image mask and scored by a forward pass — avoids gradient noise/saturation issues in deep CNNs.

Target Layer

Final DenseNet121 conv block (denseblock4 / norm5) — deep enough to encode disease-relevant spatial features, still coarse enough to localize after upsampling.

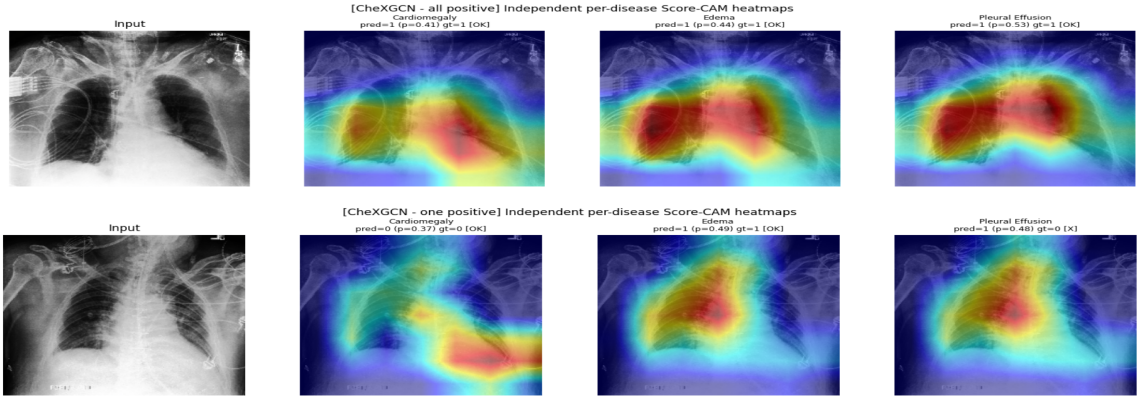
Independent Per Disease

Run separately for each of the 3 diseases using that disease's own class score — so each heatmap is independently, spatially verifiable rather than one blended saliency map.

Score-CAM — CheXGCN

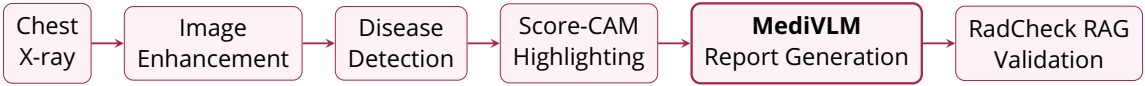
Selected for clearest per-disease spatial differentiation, independent of raw AUC/F1 ranking.

CheXGCN



MediVLM - Where It Fits in ReXTrust

Objective Generate a radiology report from a chest X-ray using region-aware visual understanding and medical language representation.

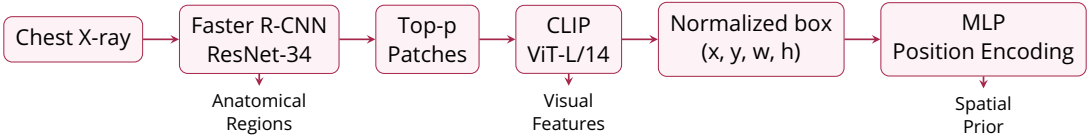


Why report generation?
Disease detection tells us what may be present. MediVLM converts visual findings into a clinical narrative.

Why MediVLM?
It focuses on salient anatomical regions and combines visual information with medical language.

Role in our system
MediVLM is the report-generation stage. RadCheck RAG is added after generation for grounding and consistency validation.

MediVLM Architecture — Visual Understanding



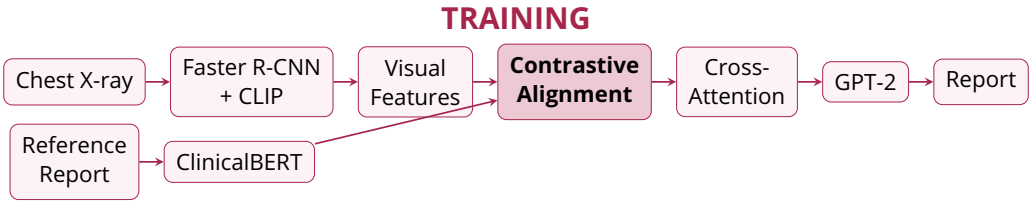
1. Detect regions
Faster R-CNN proposes salient anatomical regions using bounding boxes. It performs region extraction, not disease classification.

2. Select patches
Regions are ranked by confidence. The implementation uses the top salient patches by default.

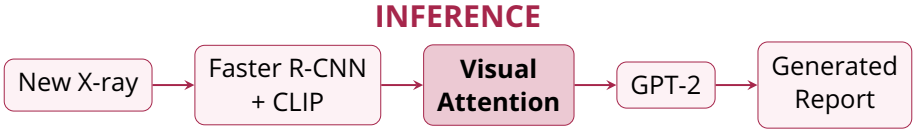
3. Add spatial context
CLIP ViT-L/14 extracts visual features. Normalized box coordinates are encoded to retain where each region occurs.

Output: visual tokens carrying appearance + spatial information.

MediVLM Architecture — Training and Inference



Training: the reference report is available, so the model learns medical text representation, image-text alignment, multimodal fusion and report generation.



Inference: no reference report is available; the new X-ray is converted to visual features and passed to the decoder to generate the report.

Datasets, Evaluation and Evidence from the Paper

Datasets

Dataset	Train	Val	Test
MIMIC-CXR	369K	3.0K	5.2K

MIMIC-CXR - large-scale chest X-ray + reports.

Training: AdamW; $LR = 2 \times 10^{-5}$; batch 32; 30-50 epochs.

Evaluation Metrics

NLG: BLEU-1-4, METEOR, ROUGE-L.
Semantic / Clinical: BERTScore, RadGraph-F1, RaTEScore.
Additional: Severity Score.

Why this architecture?

The paper reports that selective salient patching improves performance over whole-image processing, while ClinicalBERT gives stronger text representations than the alternative CLIP text encoder.

Our Results and ReXTrust Extension

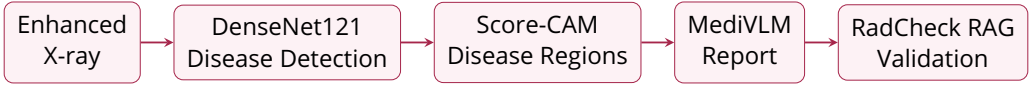
Our Report-Generation Results

Metric	GPT-2	Qwen
BLEU-1	0.3631	0.2111
BLEU-2	0.2401	0.1444
BLEU-3	0.1707	0.1061
BLEU-4	0.1228	0.0784
METEOR	0.3700	0.3456
ROUGE-L	0.2700	0.2237

Qwen = experimental alternative decoder
GPT-2 = original MediVLM decoder

Observation: GPT-2 is higher on BLEU-1-4, METEOR and ROUGE-L in our reported experiment; Qwen additionally reports BERTScore $= 0.7864$.

MediVLM inside the final ReXTrust architecture



MediVLM generates the report. RadCheck RAG validates the generated report.

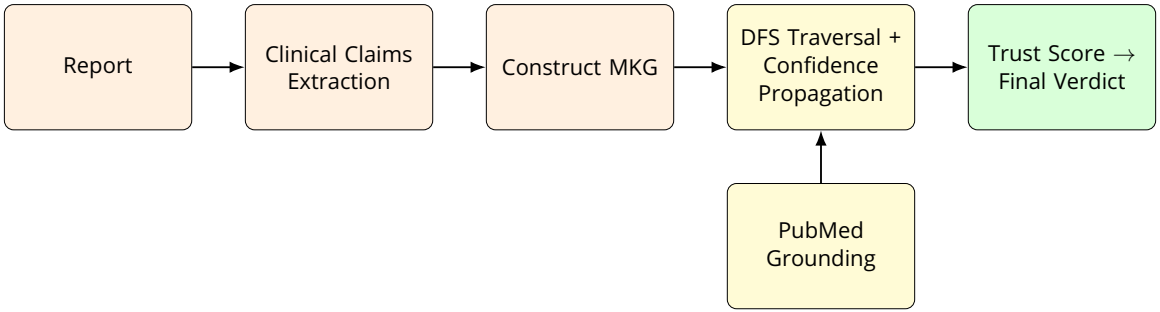
RadCheck - RAG

Clinical Hallucination Detection in Radiology Reports

Why RadCheck RAG is needed?: LLM-generated radiology reports can hallucinate findings not actually in the image — sound fluent, wrong content. RadCheck RAG fixes this.

Core Idea: Hallucination detection needs both *external* evidence (is this finding medically documented?) and *internal* consistency checks [**?**](does this contradict another finding in the same report).

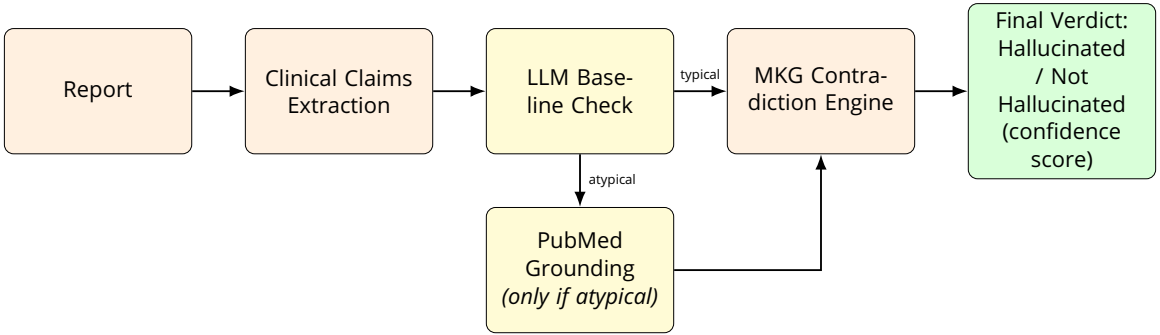
Previous Architecture



Drawbacks of Previous Architecture

- No explicit **contradiction detection** — only confidence propagation via BFS
- PubMed grounding applied **uniformly**, not selectively.
- No **bidirectional relationship modeling** between claims and KG nodes
- Lacks a structured **typical vs. atypical** decision path, increasing unnecessary computation

Architecture



Results

1. Clinical Claims Extraction

```
============================================================
  Step 1: Report Ingestion & Extraction
============================================================

  Input Report:
   "The heart is of normal size with no abnormalities. The lungs are clear with no signs of consolidation or fluid accumulation. The structures
around the lungs, including the mediastinum and hilar regions are normal. Overall, there are no acute or concerning findings."

  Extracted 4 clinical claims:

    Claim 1: "The heart is of normal size with no abnormalities"
             Target: heart | Status: normal
    Claim 2: "The lungs are clear with no signs of consolidation or fluid accumulation"
             Target: lungs | Status: clear
    Claim 3: "The structures around the lungs, including the mediastinum and hilar regions are normal"
             Target: mediastinum and hilar regions | Status: normal
    Claim 4: "no acute or concerning findings"
             Target: overall | Status: normal
```

2. LLM Baseline Check

```
============================================================
  Step 2: LLM Baseline Check (typical / atypical)
============================================================

    Claim 1: [OK] TYPICAL
      "The heart is of normal size with no abnormalities"
     Reasoning: Standard description of a normal heart size and appearance.

    Claim 2: [OK] TYPICAL
      "The lungs are clear with no signs of consolidation or fluid accumulation"
     Reasoning: Standard description of clear lungs without pathology.

    Claim 3: [OK] TYPICAL
      "The structures around the lungs, including the mediastinum and hilar regions are normal"
     Reasoning: Standard observation of normal mediastinal and hilar anatomy.

    Claim 4: [OK] TYPICAL
      "no acute or concerning findings"
     Reasoning: Common summary statement indicating a normal or non-urgent study.

  Summary: 0 atypical / 4 total findings
```

3. PubMed Grounding & MKG Contradiction Engine

```
============================================================
  Step 3: PubMed Grounding — only if atypical
============================================================

  No atypical findings detected. Skipping PubMed grounding.

============================================================
  Step 4: MKG Contradiction Engine
============================================================

  Building Medical Knowledge Graph from report claims...
    Nodes: 8 | Edges: 4
    Heart --[has_finding]--> Claim_1
    Lungs --[has_finding]--> Claim_2
    Mediastinum And Hilar Regions --[has_finding]--> Claim_3
    Overall --[has_finding]--> Claim_4

  Running contradiction detection...

  [OK] No internal contradictions detected.
```

4. Final Verdict

```
============================================================
  Step 5: Final Verdict — Hallucinated / Not Hallucinated
============================================================

    Claim 1: [PASS] NOT HALLUCINATED  (Confidence: 1.00)
      "The heart is of normal size with no abnormalities"
     Baseline: typical | PubMed Grounded: False | Contradicted: False

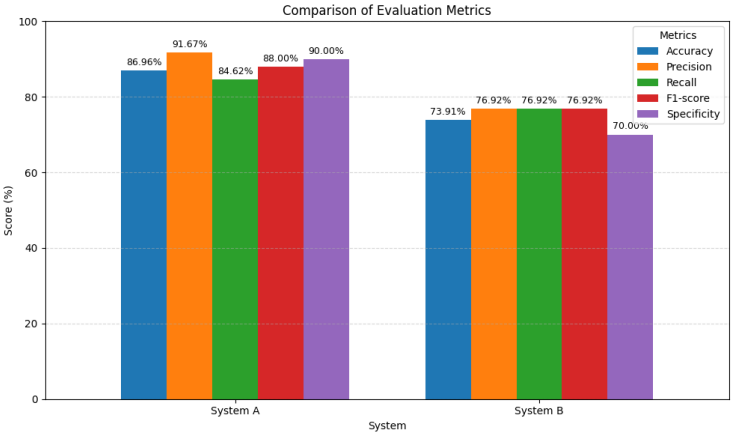
    Claim 2: [PASS] NOT HALLUCINATED  (Confidence: 1.00)
      "The lungs are clear with no signs of consolidation or fluid accumulation"
     Baseline: typical | PubMed Grounded: False | Contradicted: False

    Claim 3: [PASS] NOT HALLUCINATED  (Confidence: 1.00)
      "The structures around the lungs, including the mediastinum and hilar regions are normal"
     Baseline: typical | PubMed Grounded: False | Contradicted: False

    Claim 4: [PASS] NOT HALLUCINATED  (Confidence: 1.00)
      "no acute or concerning findings"
     Baseline: typical | PubMed Grounded: False | Contradicted: False

  ------------------------------------------------------------
  OVERALL VERDICT: [PASS] REPORT IS CLINICALLY CONSISTENT
  Average Confidence Score: 1.00
  Graph Stats: 8 entities, 4 relations
  ------------------------------------------------------------
```

Architecture Comparison



Note: Evaluation was performed on a curated set of manually labeled clinical statements rather than raw dataset reports, since MIMIC-CXR/IU-Xray reports have no ground-truth contradiction/hallucination labels needed to compute precision, recall, and F1.

Technical Enrichment — Courses

- **AI for Medical Diagnosis** – **Platform: Coursera** **(Status: Ongoing)**
- **Generative AI with LLMs** – **Platform: Coursera** **(Status: Ongoing)**
- **End-to-End Multimodal AI: Fine-Tuning, Fusion & MLOps** – **Platform: Coursera** **(Status: Ongoing)**

Team Members



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2. **Zero-DCE:** Guo et al., *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
3. **U-Net:** Ronneberger et al., *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.
4. **MediVLM:** MediVLM: A Vision Language Model for Radiology Report Generation from Medical Images.
5. **Agentic MKG:** Agentic Medical Knowledge Graphs Enhance Medical Question Answering: Bridging the Gap Between LLMs and Evolving Medical Knowledge.
6. **CheXpert:** Irvin et al., *AAAI Conference on Artificial Intelligence*, 2019.
7. **Score-CAM:** Score-CAM: Score-Weighted Visual Explanations for Convolutional Neural Networks.

Thank You